
DESIGN II

a new algorithm for designing any logic circuits, given truth table

Optimistic vs. Pessimistic minterms (SoP) vs. MAXTERMS (PoS)

a design algorithm for any digital units (logic circuits), given truth table

Y	X	$F = m_0 + m_1$
0	0	1
0	1	1
1	0	0
1	1	0

Y	X	$F=m_0+m_1$	$F'=m_2+m_3$
0	0	1	0
0	1	1	0
1	0	0	1
1	1	0	1

Y	X	$F=m_0+m_1$	$F'=m_2+m_3$	$(F')'=(m_2+m_3)'$
0	0	1	0	1
0	1	1	0	1
1	0	0	1	0
1	1	0	1	0



Augustus De Morgan
(1806–1871)

Mathematician

Logician

(you see how logic helped cs)

DE MORGAN'S LAWS

▶ $(YX)' = Y' + X'$

▶ $(Y + X)' = Y'X'$

Y	X	$F = m_0 + m_1$	$F' = m_2 + m_3$	$F = (m_2 + m_3)'$	$F = m_2' m_3'$
0	0	1	0	1	1
0	1	1	0	1	1
1	0	0	1	0	0
1	1	0	1	0	0

MAXTERM

aka. Standard Sum (ORs)

X' vs. X

1 binary variable appear either:

- in its normal form X , or
- in its complement form X'

$$M_0 = m'_0$$

$$(X')' = X$$

$$M_1 = m'_1$$

$$X'$$

$Y + X$ vs. $Y + X'$ vs. $Y' + X$ vs. $Y' + X'$

2 binary variables appear either in one of these forms:



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DE MORGAN'S LAWS

▶ $(YX)' = Y' + X'$

▶ $(Y + X)' = Y'X'$

$$M_0 = m'_0$$

$$M_1 = m'_1$$

$$M_2 = m'_2$$

$$M_3 = m'_3$$

$$(Y'X')' = Y + X$$

$$(Y'X)' = Y + X'$$

$$(YX')' = Y' + X$$

$$(YX)' = Y' + X'$$

$Z + Y + X$ vs. $Z + Y + X'$ vs. ...

3 binary variables appear either in one of these forms: how many?

$Z + Y + X$ vs. $Z + Y + X'$ vs. ...

3 binary variables appear either in one of these forms: how many?

Each variable can take 2 forms (normal and complement)

We have 3 variables, $2 \times 2 \times 2 = 2^3 = 8$

$M_0 = m'_0$	$(Z'Y'X')' = Z + Y + X$
$M_1 = m'_1$	$(Z'Y'X)' = Z + Y + X'$
$M_2 = m'_2$	$(Z'YX')' = Z + Y' + X$
$M_3 = m'_3$	$(Z'YX)' = Z + Y' + X'$
$M_4 = m'_4$	$(ZY'X')' = Z' + Y + X$
$M_5 = m'_5$	$(ZY'X)' = Z' + Y + X'$
$M_6 = m'_6$	$(ZYX')' = Z' + Y' + X$
$M_7 = m'_7$	$(ZYX)' = Z' + Y' + X'$

$$A_n + \dots + A_2 + A_1 + A_0 \text{ vs. } A_n + \dots + A_2 + A_1 + A'_0 \dots$$

n binary variables appear either in one of these forms: how many?

Each variable can take 2 forms (normal and complement)

We have n variables, $2 \times 2 \times 2 \times \dots \times 2 = 2^n$

$M_0 = m'_0$	$A_n + A_2 + A_1 + A_0$
$M_1 = m'_1$	$A_n + \cdots A_2 + A_1 + A'_0$
$M_2 = m'_2$	$A_n + \cdots A_2 + A'_1 + A_0$
$\vdots$	$\vdots$
$\vdots$	$\vdots$
$M_{2^n-3} = m'_{2^n-3}$	$A'_n + \cdots A'_2 + A_1 + A'_0$
$M_{2^n-2} = m'_{2^n-2}$	$A'_n + \cdots A'_2 + A'_1 + A_0$
$M_{2^n-1} = m'_{2^n-1}$	$A'_n + \cdots A'_2 + A'_1 + A'_0$

TRUTH TABLE

en.wikipedia.org/wiki/Truth_table

TRUTH TABLE $\leftrightarrow$ MAXTERM

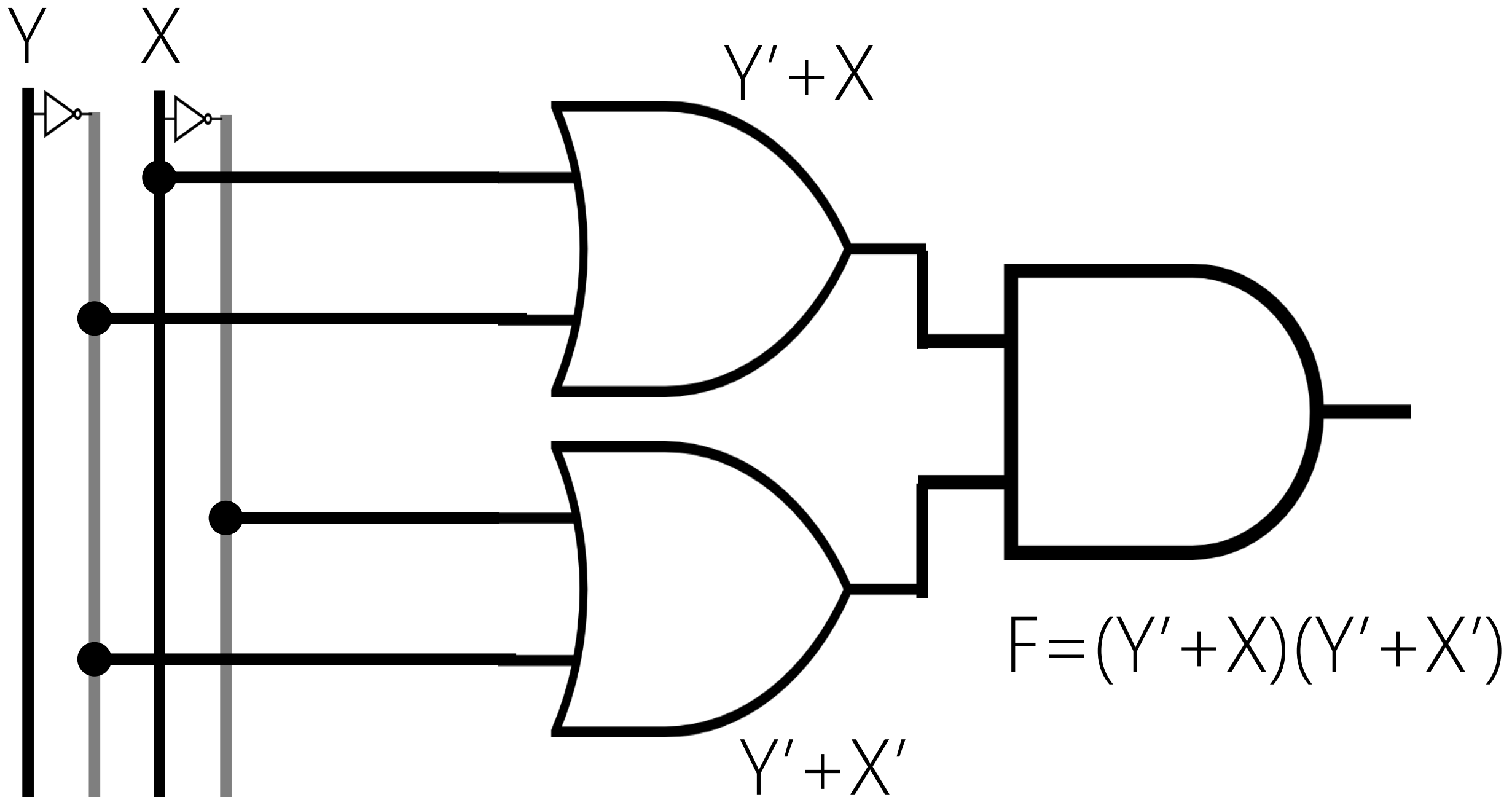
Y	X	$F = m_0 + m_1$
0	0	1
0	1	1
1	0	0
1	1	0

Y	X	$F = (F')' = m'_2 m'_3 = M_2 M_3$
0	0	1
0	1	1
1	0	0
1	1	0

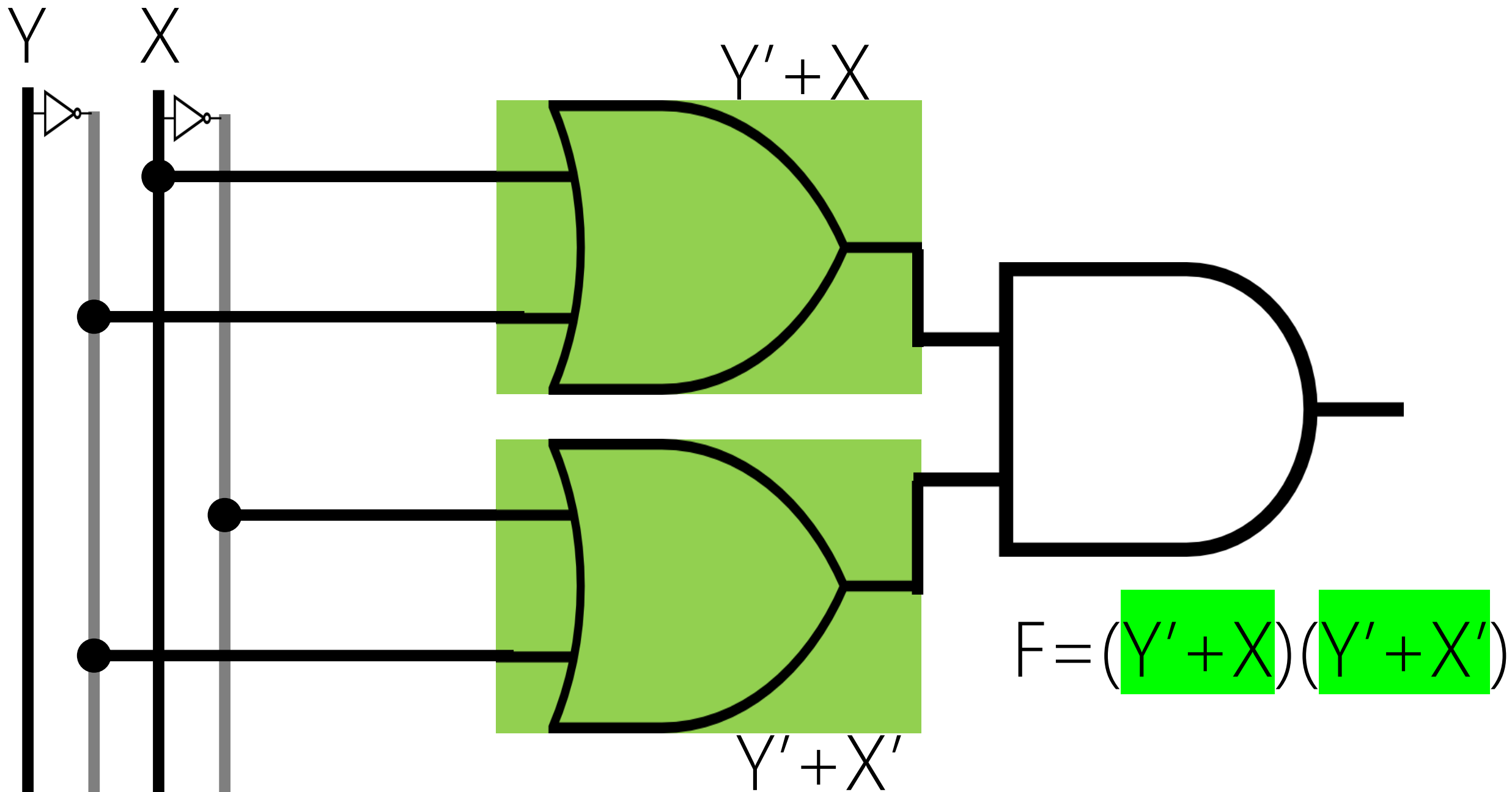
Y	X	$F = (F')' = m'_2 m'_3 = M_2 M_3$
0	0	1
0	1	1
1	0	0
1	1	0

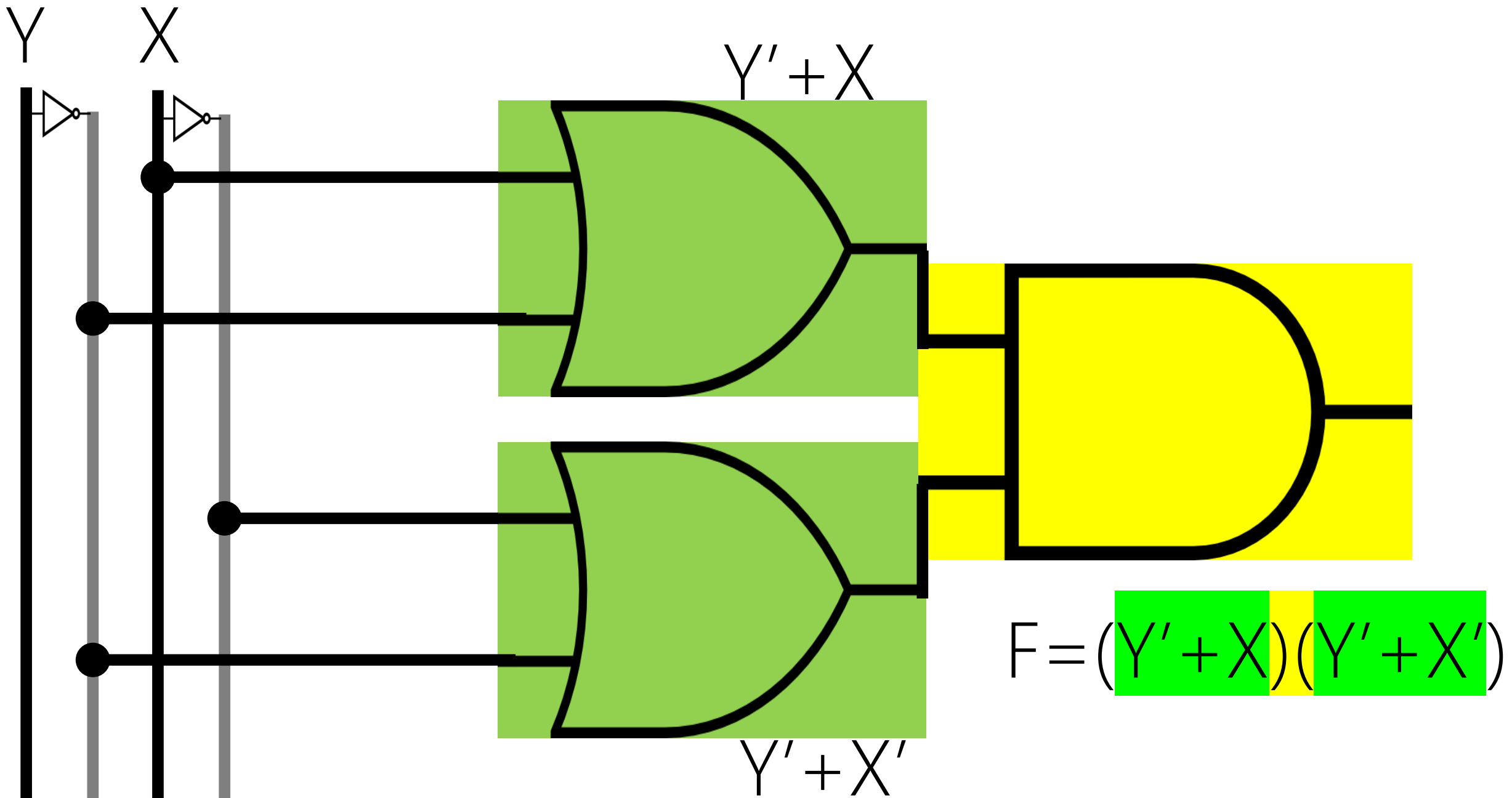
Y	X	$F = (F')' = M_2 M_3 = \prod M(2,3)$
0	0	1
0	1	1
1	0	0
1	1	0

Y	X	$F = \prod M(2,3) = (Y' + X)(Y' + X')$
0	0	1
0	1	1
1	0	0
1	1	0

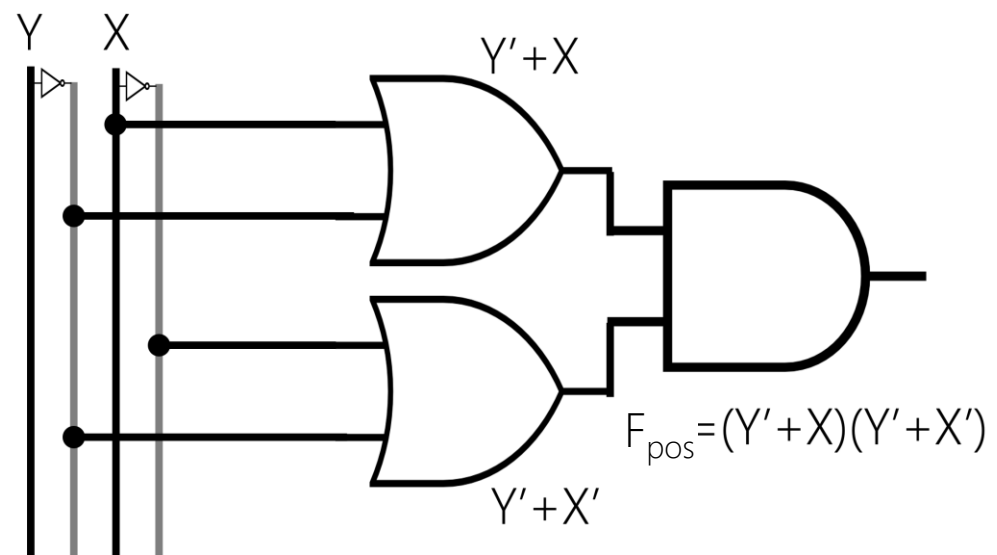
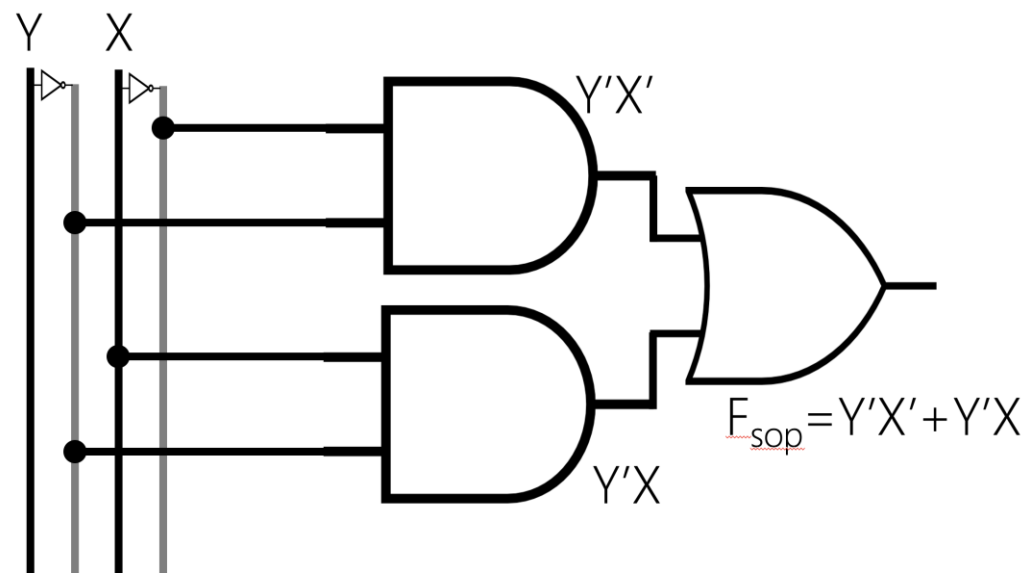


PRODUCT OF SUMS (POS)





2 LEVELS
OR → AND



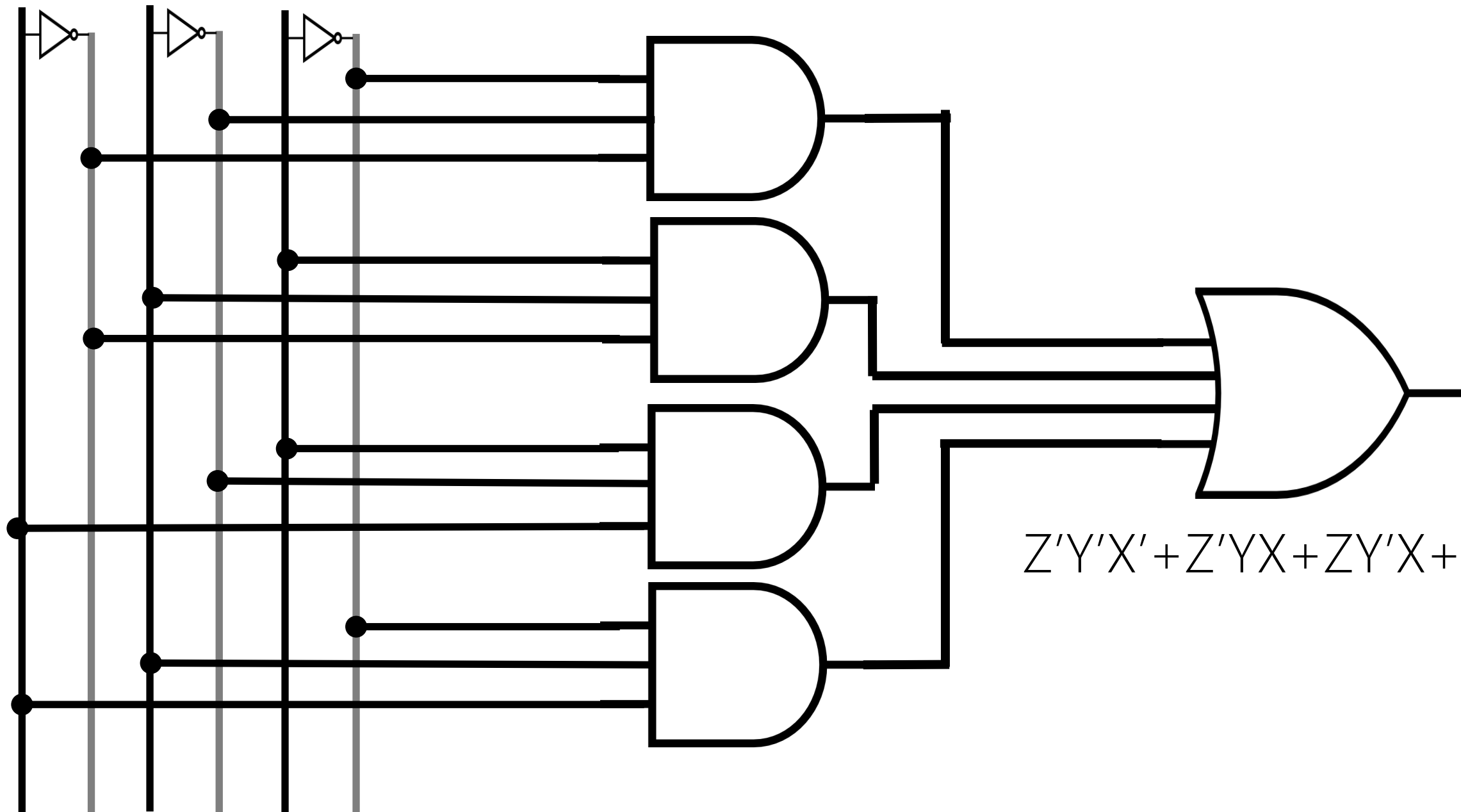
Given 3 inputs, design a circuit to
determine if there is even
number of 1

Z	Y	X	F(Z,Y,X)=?
0	0	0	?
0	0	1	?
0	1	0	?
0	1	1	?
1	0	0	?
1	0	1	?
1	1	0	?
1	1	1	?

Z	Y	X	F(Z,Y,X)=?
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	F(Z,Y,X)=Σm(0,3,5,6)
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z Y X



$$Z'Y'X' + Z'YX + ZY'X + ZYX'$$

Z	Y	X	$F(Z,Y,X)=M_1$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

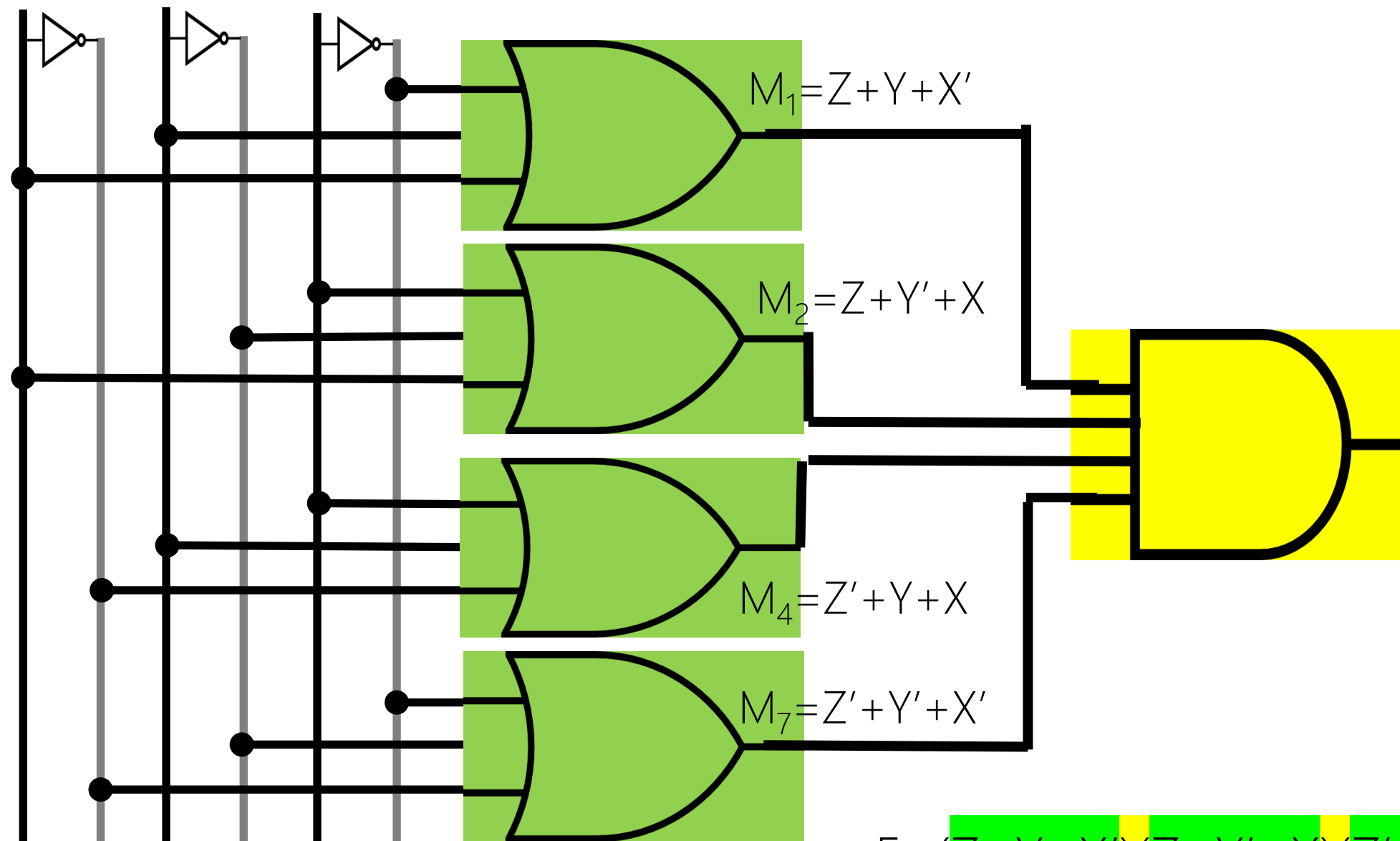
Z	Y	X	F(Z,Y,X)=M ₁ M ₂
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=M_1M_2M_4$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=M_1M_2M_4M_7$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=M_1M_2M_4M_7=\prod M(1,2,4,7)$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z Y X



$$F = (Z + Y + X')(Z + Y' + X)(Z' + Y + X)(Z' + Y' + X')$$

DESIGN I vs. II
SoP vs. PoS

Lecture Assignment